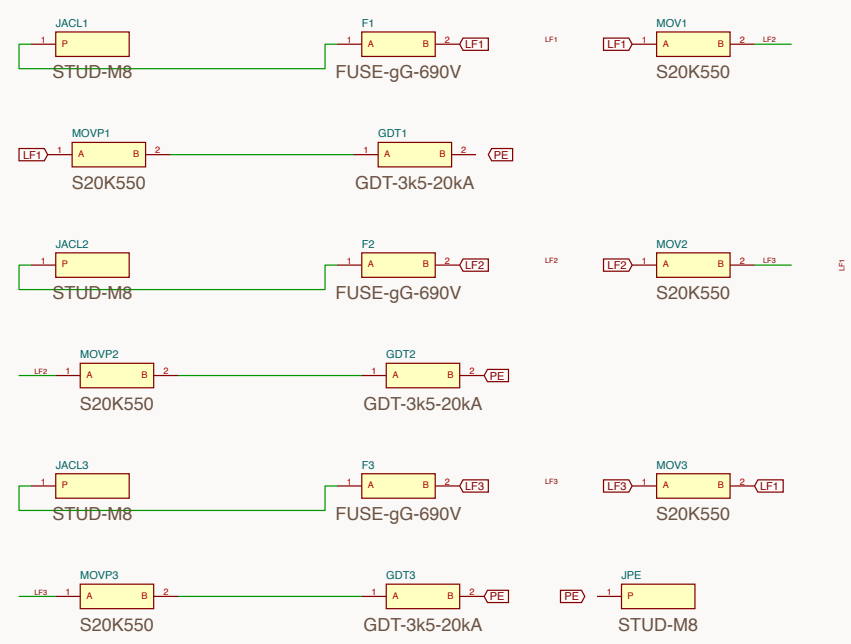
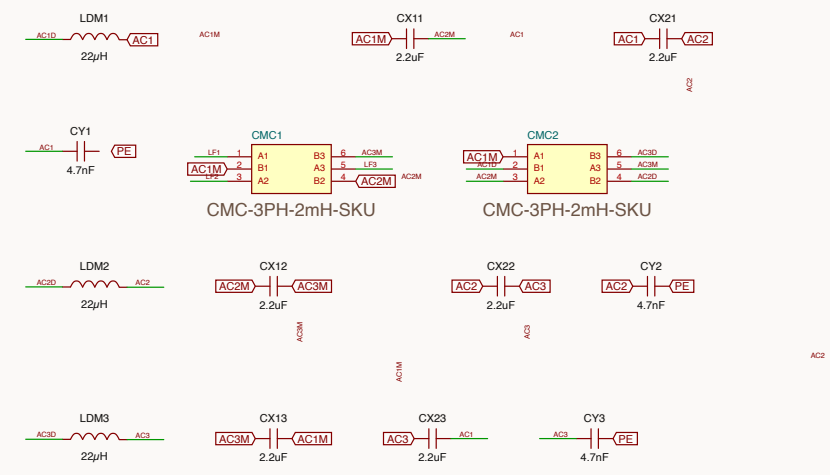


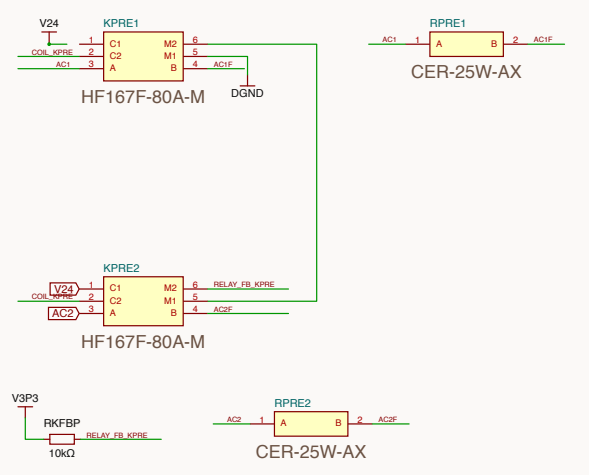
AC INPUT & PROTECTION



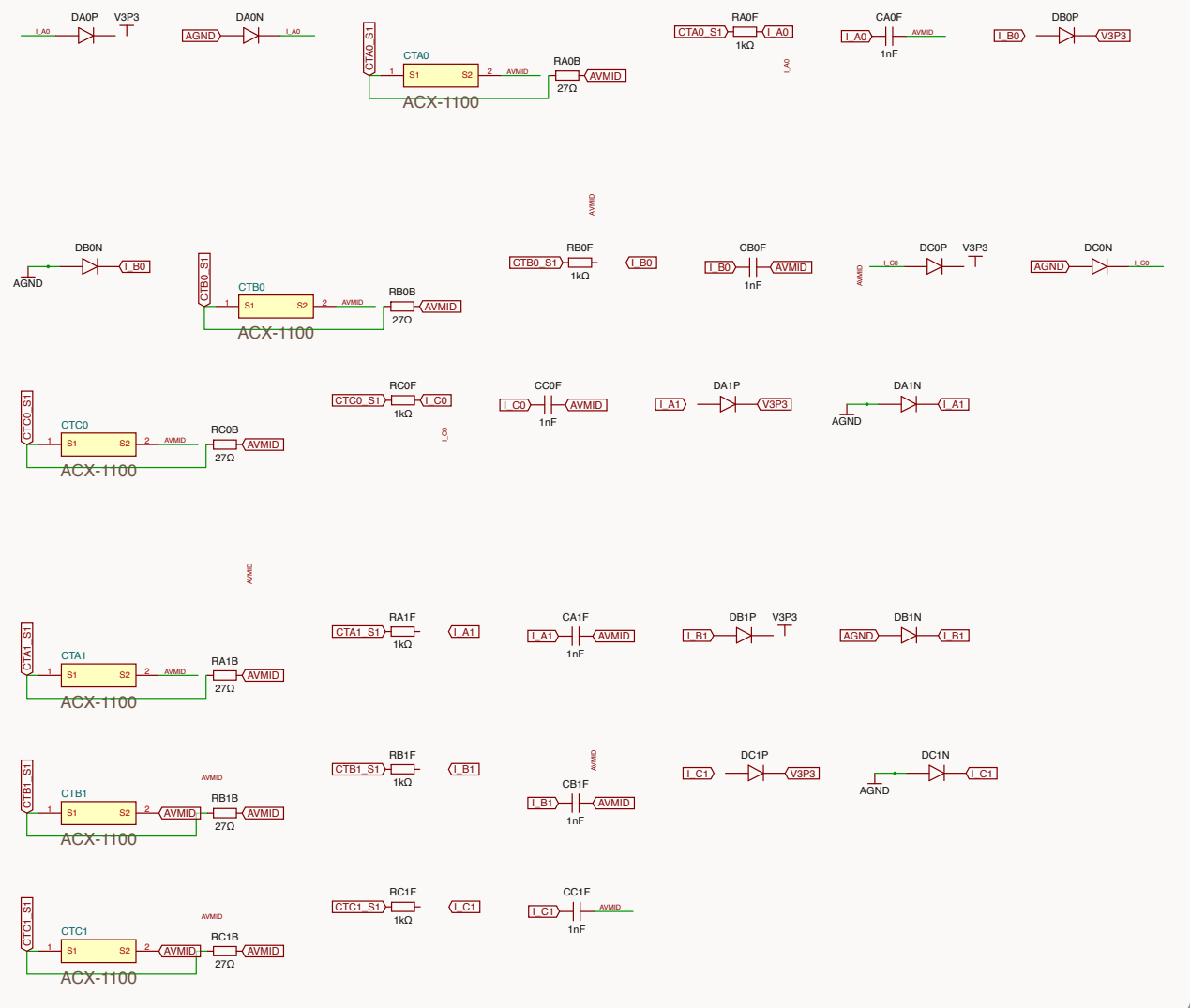
EMI FILTER



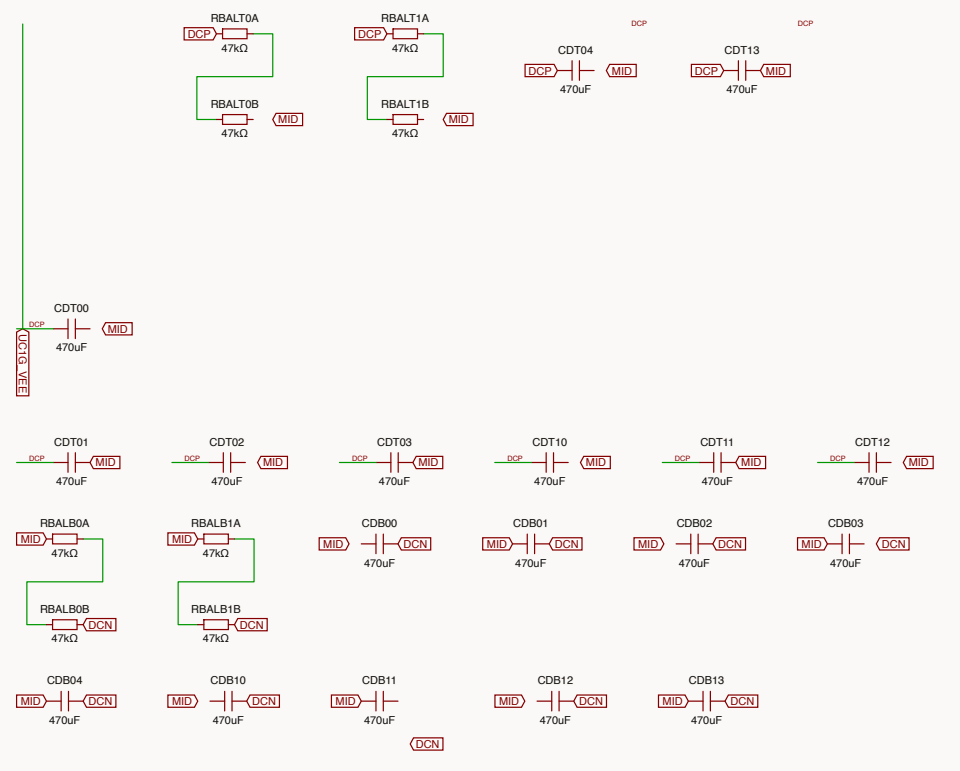
PRECHARGE



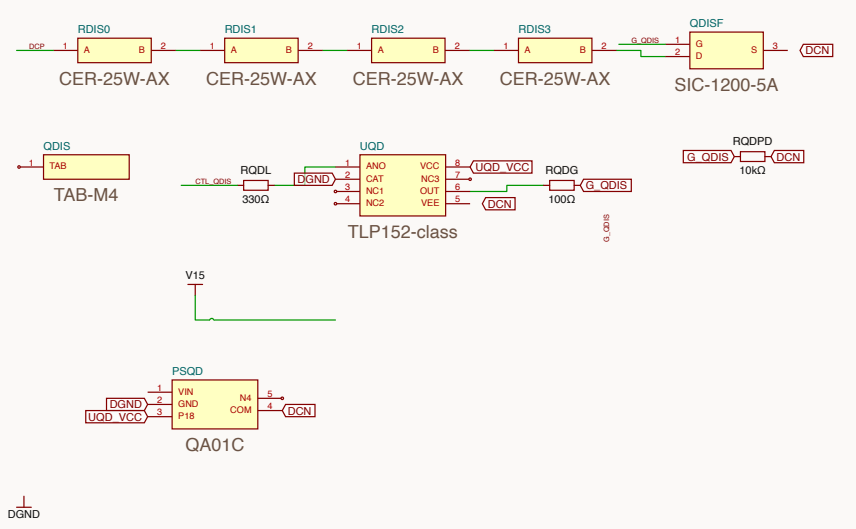
LINE CTs



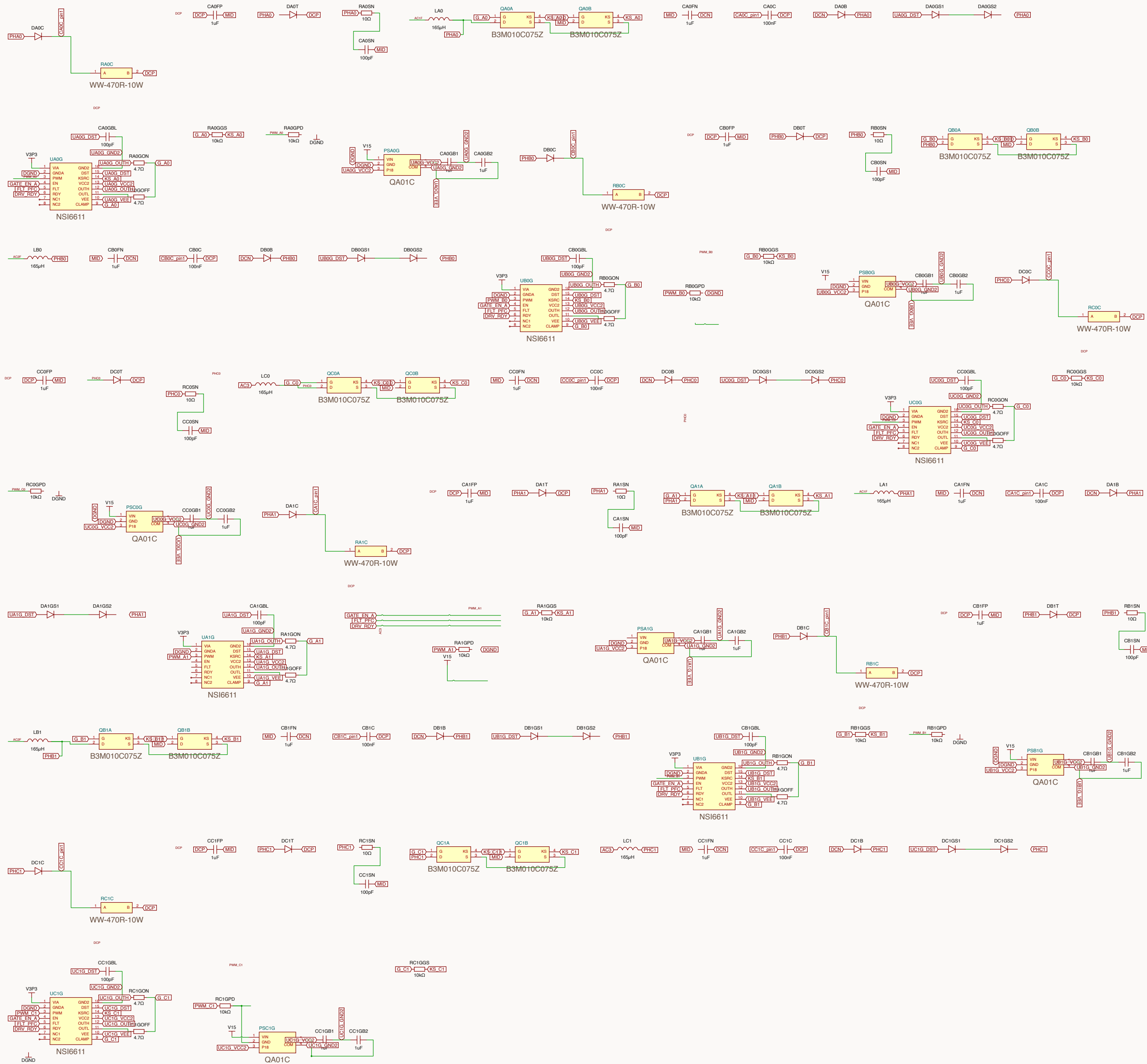
DC LINK



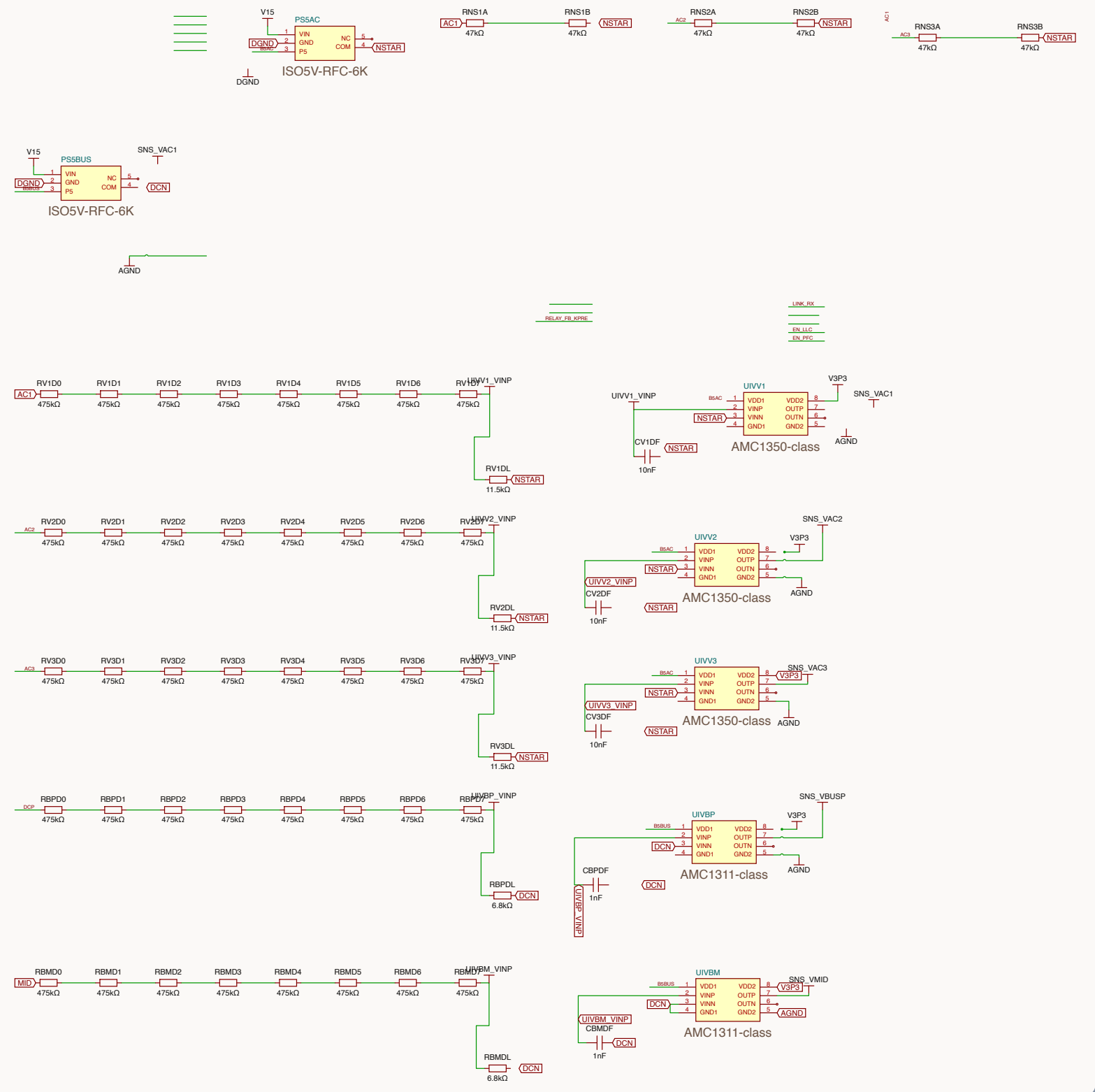
BUS DISCHARGE



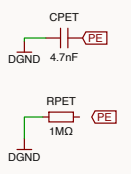
VIENNA PFC LANES



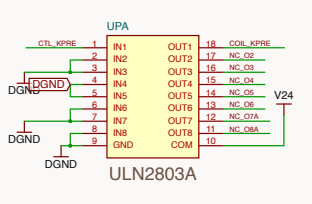
ISOLATED HV SENSING



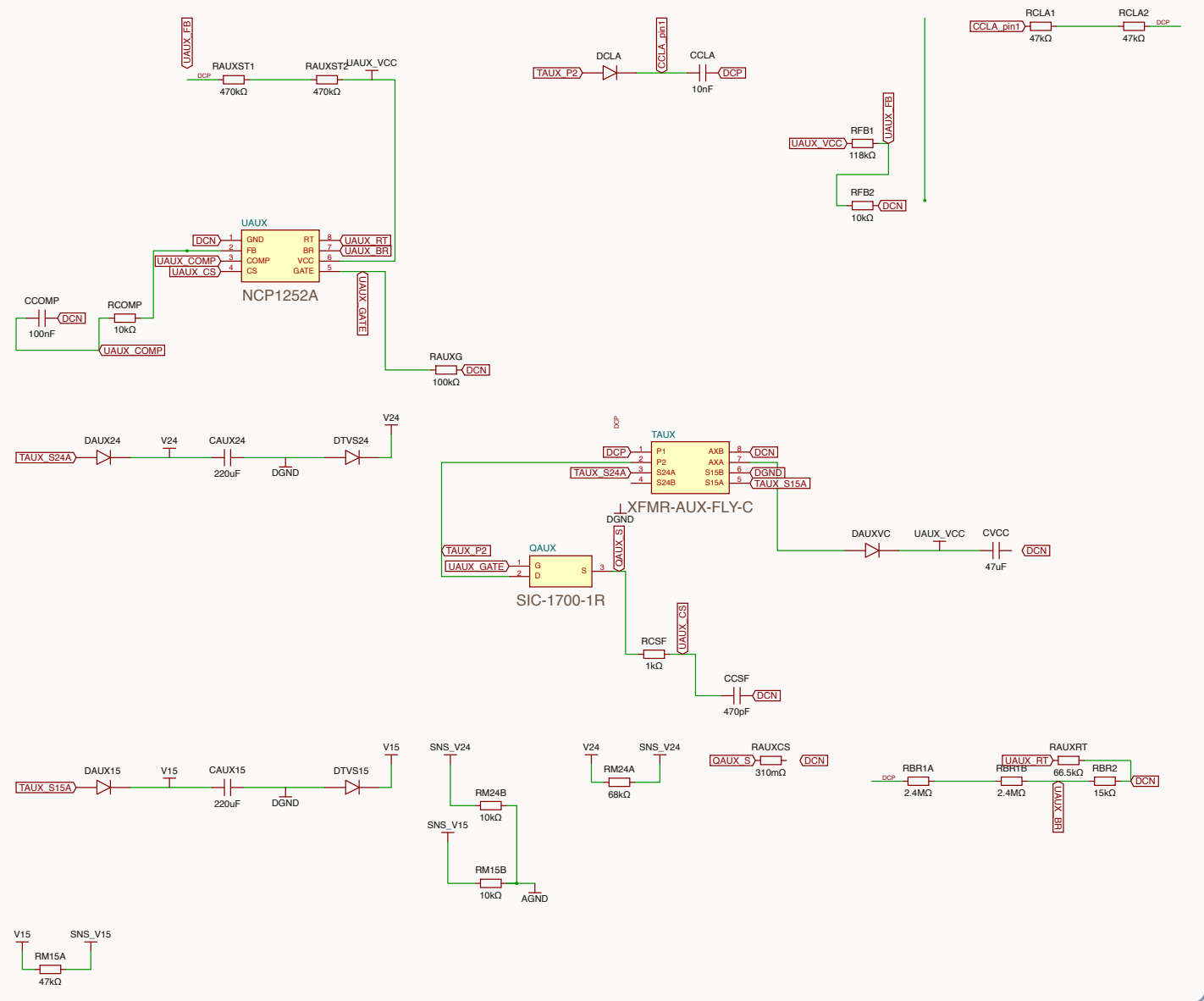
GROUND BONDS



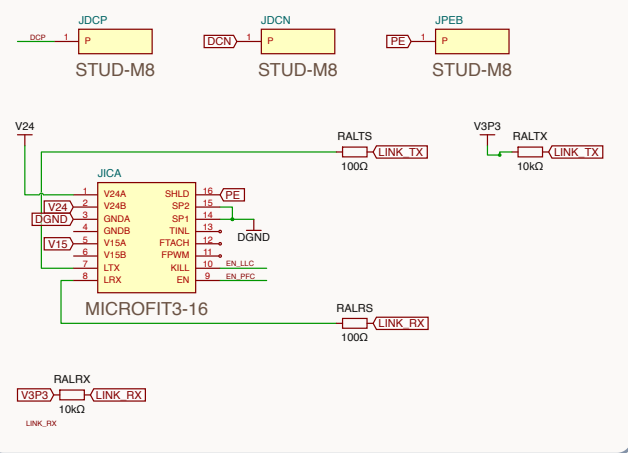
COIL DRIVER



AUX 110 W FLYBACK + 3V3 + RAIL MONITORS



DC OUT + HARNESS



The image displays a 4x4 grid of 16 circuit diagrams, each representing a different configuration for the SG2M023120LJ component. Each diagram includes a central component labeled SG2M023120LJ, which is a 120V, 0.5A, 120W power MOSFET. The diagrams are arranged in a grid, with each row and column showing a different set of connections and components. The components are labeled with NSI6611 and SG2M023120LJ. The diagrams show various input and output pins connected to different components, including diodes, capacitors, and other integrated circuits. The diagrams are labeled with NSI6611 and SG2M023120LJ.

The diagram illustrates the CRISPR-Cas9 system and its application in genome editing. It is divided into two main parts: the CRISPR-Cas9 system and its application in genome editing.

CRISPR-Cas9 System:

- CRISPR Array:** A series of DNA segments (CRISPR units) flanked by direct repeats (DR) and spacers (S). The units are transcribed into a long crRNA, which is then processed into individual crRNAs.
- Cas9 Gene:** A gene encoding the Cas9 protein, which is transcribed and translated into the Cas9 protein.
- Tracrse and crRNA Genes:** Genes encoding the tracrse and crRNA genes, which are transcribed and translated into the tracrse and crRNA proteins.

Application in Genome Editing:

- Target Identification:** A target sequence in the genome is identified, flanked by DR and S regions.
- CRISPR Array Construction:** The target sequence is integrated into the CRISPR array, creating a new crRNA.
- CRISPR Array Transcription:** The CRISPR array is transcribed into a long crRNA, which is then processed into individual crRNAs.
- CRISPR Array Processing:** The crRNA is processed into individual crRNAs, which are then loaded into the Cas9 protein.
- CRISPR Array Editing:** The Cas9 protein, guided by the crRNA, targets the specific sequence in the genome for editing.

[illegible]

The figure displays two genetic maps of the CER-2 locus on chromosome 2L in *Drosophila melanogaster*. The top map shows the CER-2 locus with genes RPVLA, RPVLA, VOM121T, RPVLA, and CER2A. The bottom map shows the CER-2 locus with genes RPVLB, VOM121T, RPVBB, and CER2B. The maps are color-coded to show different alleles and their positions relative to the CER-2 locus.

Top Map: CER-2-10W-AX CER-2-10W-AX CER-2-10W-AX CER-2-10W-AX

Genes: RPVLA, RPVLA, VOM121T, RPVLA, CER2A

Alleles: CER-2-10W-AX (red), CER-2-10W-AX (green), CER-2-10W-AX (blue), CER-2-10W-AX (yellow)

Bottom Map: CER-2-10W-AX CER-2-10W-AX CER-2-10W-AX CER-2-10W-AX

Genes: RPVLB, VOM121T, RPVBB, CER2B

Alleles: CER-2-10W-AX (red), CER-2-10W-AX (green), CER-2-10W-AX (blue), CER-2-10W-AX (yellow)

The schematic representation shows the following constructs and their components:

- VOP3**: Promoter (green line) - RTILLOP (yellow box) - Terminator (red line)
- RTILLOP**: Promoter (green line) - RTILLOP (yellow box) - Terminator (red line)
- JTLIC**: Promoter (green line) - JTLIC (yellow box) - Terminator (red line)
- PH-2**: Promoter (green line) - PH-2 (yellow box) - Terminator (red line)
- JTXFRP**: Promoter (green line) - JTXFRP (yellow box) - Terminator (red line)
- PH-2**: Promoter (green line) - PH-2 (yellow box) - Terminator (red line)
- JTXFRP**: Promoter (green line) - JTXFRP (yellow box) - Terminator (red line)

Each construct is shown with its components: a promoter (green line), a coding region (yellow box), and a terminator (red line). The constructs are labeled with their respective names and the 100bp scale bar.

[illegible]

The diagram shows the Microfit32-16 board with the following connections:

- 16C45:**
 - VDD to +5V
 - VSS to GND
 - PBLTX to 15C42
- 16C46:**
 - VDD to +5V
 - VSS to GND
 - PBLRX to 15C43
 - 15C44 to 15C43

[illegible]

The schematic diagram illustrates the LED driver circuit. It features a 74HC595 shift register at the core, which is powered by a +5V supply and ground. The shift register's Q0-Q7 outputs are connected to the DS90C03 differential line drivers. These drivers are connected to the DS90C01 differential line receivers, which in turn drive the S8050 transistors. The transistors are connected to the LED array (not shown) and ground. The circuit is powered by a +5V supply and ground.

60 kW module · DC-DC (upper) board · sheet 2 of 3 · 2x Vienna PFC cells + 2x 3-ph LLC cells
14 sections · 238 wired clusters · 425 symbols · cross-section links are net labels

generated 2026-09-08
LCSC: calculations/out/bom-60kw.csv

[illegible]

JCARD

Pin	Function	Pin	Function
1	PWM0	68	PWM0
2	PWM1	69	PWM1
3	PWM2	70	PWM2
4	PWM3	71	PWM3
5	PWM4	72	PWM4
6	PWM5	73	PWM5
7	PWM6	74	PWM6
8	PWM7	75	PWM7
9	PWM8	76	PWM8
10	PWM9	77	PWM9
11	PWM10	78	PWM10
12	PWM11	79	PWM11
13	AIN0	80	AIN0
14	AIN1	81	AIN1
15	AIN2	82	AIN2
16	AIN3	83	AIN3
17	AIN4	84	AIN4
18	AIN5	85	AIN5
19	AIN6	86	AIN6
20	AIN7	87	AIN7
21	AIN8	88	AIN8
22	AIN9	89	AIN9
23	AIN10	90	AIN10
24	AIN11	91	AIN11
25	AIN12	92	AIN12
26	TSNS0	93	TSNS0
27	TSNS1	94	TSNS1
28	AVMMID	95	AVMMID
29	AGND_2	96	AGND
30	DO0	97	DO0
31	DO1	98	DO1
32	DO2	99	DO2
33	DO3	100	DO3
34	DO4	101	DO4
35	DO5	102	DO5
36	DO6	103	DO6
37	DIO	104	DIO
38	DIT	105	DIT
39	DI2	106	DI2
40	DI3	107	DI3
41	DI4	108	DI4
42	DI5	109	DI5
43	GATE_EN	110	GATE_EN
44	FLT	111	FLT

CONN-CARD-88

[illegible][illegible]

V3P3

RROLE0

10kΩ

ROLE0

RROLE1

V3P3

10kΩ

ROLE1

DC-Modules — Control Card (GD32G553VET6, one card for both converter roles)	Rev D.1
30/60 kW module · Control card board · sheet 3 of 3 · role-agnostic: AC-DC or DC-DC slot via ROLE straps 8 sections · 37 wired clusters · 45 symbols · cross-section links are net labels	generated 2026-09-08 LCSC: calculations/out/bom-control-card.csv